

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	18 June 2026
Team ID	
Project Name	BOOK A DOCTOR
Maximum Marks	4 Marks

Technical Architecture:

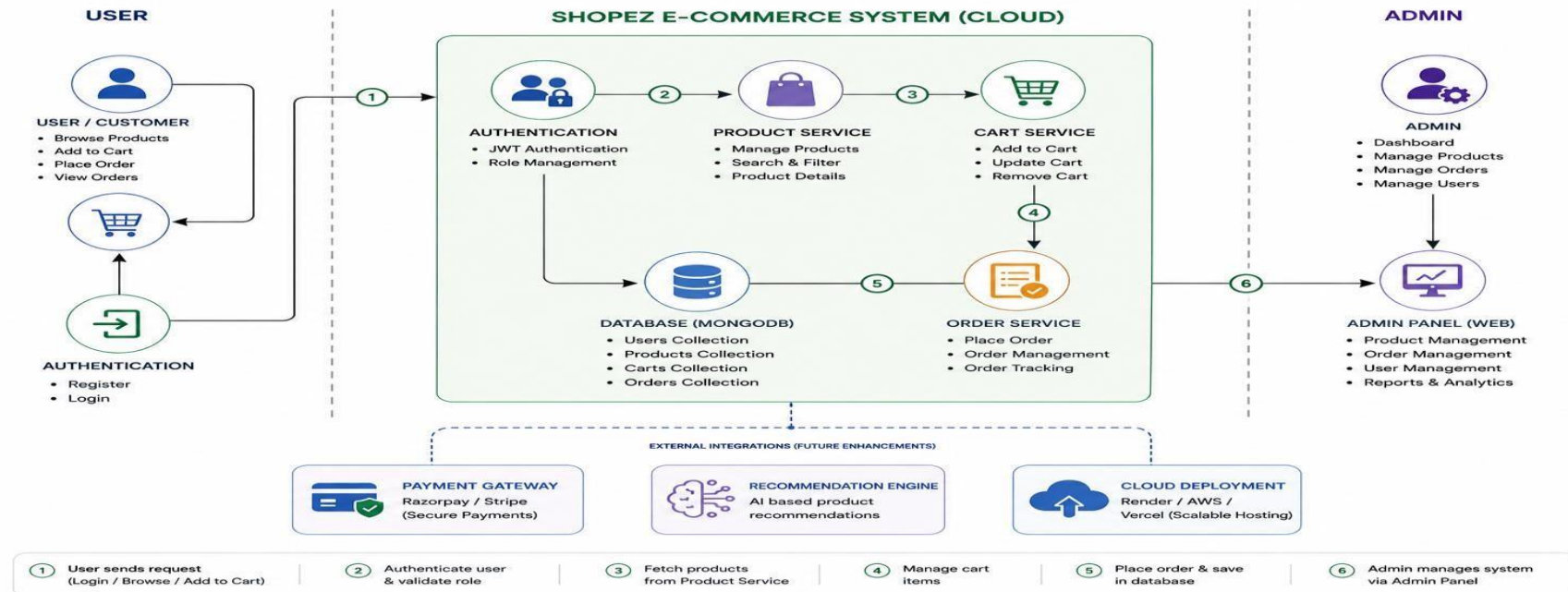


Table-1 : Components

S.No	Component	Description	Technology
1	User Interface	Web application for patients, doctors, and administrators	React.js, HTML5, CSS3, JavaScript, Bootstrap
2	Authentication	User registration, login, and role-based access	Node.js, Express.js, JWT, bcrypt
3	Doctor Management	View doctor profiles, specialties, and availability	React.js, Node.js, Express.js
4	Appointment Management	Book, update, cancel, and manage appointments	Node.js, Express.js, MongoDB
5	Database	Stores users, doctors, appointments, and reports	MongoDB
6	Cloud Database	Cloud-hosted NoSQL database service	MongoDB Atlas
7	File Storage	Upload prescriptions and medical reports securely	Multer, Local Storage / Cloud Storage
8	Notification Service	Sends appointment confirmations and reminders	Nodemailer, Gmail SMTP
9	REST API	Handles frontend-backend communication	Express.js REST API, Axios
10	Security Layer	Authentication, authorization, and password encryption	JWT, bcrypt, HTTPS
11	Deployment	Production hosting and deployment	Render, Vercel, AWS

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used in development	React.js, Node.js, Express.js, MongoDB
2	Security Implementations	Authentication, authorization and password protection	JWT Authentication, Bcrypt Password Hashing, HTTPS, OWASP Security Practices
3	Scalable Architecture	Three-tier architecture separating frontend, backend and database	React.js + Express.js + MongoDB
4	Availability	Cloud deployment with high availability and backup support	MongoDB Atlas, Render/Vercel
5	Performance	Fast API responses, optimized database queries and caching support	MongoDB Indexing, REST APIs, React Virtual DOM

